Certainty Thermometer Student Handout

TOK Cognitive Science Activity Bank • Cross-AOK • 25-35 min

Category	Cross-AOK	**Time**	25-35 min
Difficulty	Easy	**ID**	certainty-thermometer

Big idea: Certainty is not one simple measure; it depends on evidence, method, context, and purpose.

Student-Facing Prompt

Inspect Certainty Thermometer Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Certainty Thermometer Stimulus A	Student-facing stimulus 1: Claim cards: '2+2=4', 'This source is reliable', 'This song is sad', 'The medicine works', 'The policy is fair'. Name the AOK, the method being used, and the standard of evidence you would apply.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Certainty Thermometer Stimulus B	Student-facing stimulus 2: Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes. Name the AOK, the method being used, and the standard of evidence you would apply.	Use this for comparison, a second round, or a partner challenge.
Certainty Thermometer Reveal / Twist	Student-facing stimulus 3: Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes. Name the AOK, the method being used, and the standard of evidence you would apply.	Teacher reveal: connect the changed response to the big idea — Certainty is not one simple measure; it depends on evidence, method, context, and purpose.

Actual Lesson Questions

#	Question for students
1	After inspecting Certainty Thermometer Stimulus A, what is your first knowledge claim?
2	Which exact detail from Certainty Thermometer Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Is certainty the best measure of knowledge?

Ready-to-Use Examples

• Claim cards: '2+2=4', 'This source is reliable', 'This song is sad', 'The medicine works', 'The policy is fair'.

• Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Cross-AOK.

Activity Steps

1. Give groups claim cards and ask them to place each on a certainty thermometer.
2. Groups explain the criterion they used first.
3. Introduce criteria cards such as proof, replication, interpretation, consensus, usefulness, and ethical stakes.
4. Students revise placements using at least two criteria.
5. Debrief why certainty differs across knowledge contexts.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

• Knowledge question: Is certainty the best measure of knowledge?

• Can less certain knowledge be more useful or meaningful?

• How should knowers compare standards across AOKs?

Knowledge Questions

• Is certainty the best measure of knowledge?

• How should standards be compared across AOKs?

• Can uncertainty be a strength in some knowledge contexts?

Transfer Example

For example, run this activity with claims placed on a scale from guess to justified certainty. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...